

Public Economics

Lecture 4 · Taxation

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Today's Questions

- What are the different types of taxes governments levy, and how does the burden of a tax actually get shared between buyers and sellers?
- What does it mean for a tax system to be “fair,” and how do we measure that?

Based on

Gruber, J. *Public Finance and Public Policy*, Ch. 18

4.1 Types of Taxation

4.2 Tax Incidence

4.3 Measuring the Fairness of Tax Systems

4.4 Defining the Income Tax Base

The Rise of the Fiscal State

4.5 The Appropriate Unit of Taxation

The Distribution of the Tax Burden

4.1 Types of Taxation

Five Types of Taxes

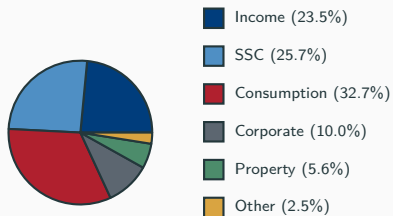
- **Payroll tax:** levied on earnings from work; finances social insurance (pensions, unemployment, health).
- **Individual income tax:** paid on income accrued during the year — a broader base than payroll (interest, etc.) and often assessed per household, not per worker.
- **Corporate income tax:** levied on corporate earnings, to catch capital income that might otherwise escape individual-level taxation.
- **Wealth taxes:** levied on the *value of assets* held (not income), e.g. property tax, estate/inheritance tax.
- **Consumption taxes:** levied on spending, e.g. VAT/sales tax; when applied to one specific good (fuel, tobacco, alcohol), called an **excise tax**.

The Same Five Types, in Austria

Type	Austrian tax(es)
Payroll	<i>Sozialversicherungsbeiträge</i> — pension, health, unemployment, accident insurance; split employer/employee
Individual income	<i>Einkommensteuer</i> (self-employment/business income); <i>Lohnsteuer</i> (withholding on wage income)
Corporate income	<i>Körperschaftsteuer</i> (KSt)
Wealth	<i>Grundsteuer</i> (property) and <i>Grunderwerbsteuer</i> (real-estate transfer) still exist; <i>Vermögensteuer</i> and <i>Erbschafts-/Schenkungssteuer</i> do not
Consumption	<i>Umsatzsteuer</i> (VAT); excise: <i>Mineralölsteuer</i> (fuel), <i>Tabaksteuer</i> , <i>Alkohol-/Biersteuer</i> , <i>NoVA</i> (vehicles), <i>CO₂-Bepreisung</i> (since 2022)

International Tax Revenues by Type

OECD average



Taxation Around the World: Tax Revenue Composition

18.1

CHAPTER 18 ■ TAXATION IN THE UNITED STATES AND AROUND THE WORLD

Taxation Around the World

	Norway	Denmark	OECD Average
Individual income taxes	24%	55%	25%
Social insurance contributions (payroll tax)	23	2	27
Corporate taxes	22	5	8
Consumption tax	26	30	31
Property tax	3	4	5
Other	2	4	4

Two very different high-tax countries: Norway spreads revenue broadly across income, payroll, and corporate taxes; Denmark relies overwhelmingly on the individual income tax alone.

Source: Gruber, *Public Finance and Public Policy*, §18.1, "Taxation Around the World."

Where Does Austria Fit?

- Austria's tax mix leans heavily on **labor**
- Austria abolished its general **wealth tax** (*Vermögensteuer*) in 1994, and its **inheritance/gift tax** lapsed in 2008 after the *Verfassungsgerichtshof* ruled the property-valuation method unconstitutional and no replacement was passed.
- Austria has had *no* wealth or inheritance tax since.

4.2 Tax Incidence

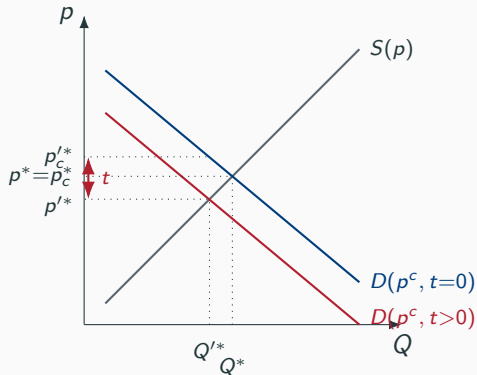
Tax Incidence: Who Really Pays?

Tax incidence

Tax incidence measures who bears the legal or economic burden of a tax.

- **Legal (statutory) incidence:** who is legally required to send the tax payment to the government.
- **Economic incidence:** who actually bears the burden, once prices adjust.

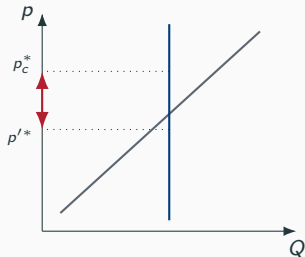
The Canonical Case: An Excise Tax



A tax t on consumers wedges $p_c = p + t$ apart from p . Quantity falls from Q^* to Q'^* ; the consumer price rises by *less than* t — both sides share the burden.

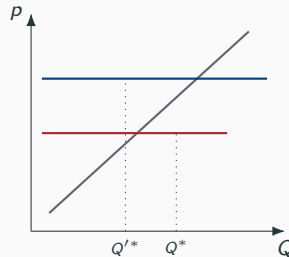
Two Extreme Cases

Perfectly inelastic demand



Quantity can't change — consumers bear **100%** of the burden.

Perfectly elastic demand



Consumer price can't change — producers bear **100%** of the burden.

The side of the market with the *less elastic* response bears more of the tax — regardless of who writes the check.

Elasticity Determines Incidence

- In theory, incidence does *not* depend on who is statutorily required to pay — a tax on sellers and an equivalent tax on buyers have the same economic incidence.
- What matters is the **relative elasticity of supply and demand**:

$$\varepsilon_D = \frac{p_c}{D} \frac{dD}{dp_c}, \quad \varepsilon_S = \frac{p}{S} \frac{dS}{dp}$$

Application: excise taxes in Austria

Austria taxes tobacco, alcohol, mineral oil, and (since 2022) CO₂ emissions via excise duties. Demand for cigarettes and heating fuel is famously **inelastic** in the short run — so standard incidence theory predicts these taxes fall mostly on *consumers*, not producers.

The Equity–Efficiency Trade-off, Revisited

A progressive income tax has two effects:

1. **Equity gain:** reduces income inequality.
2. **Efficiency loss:** people may work less when income is taxed — if they care about *net*, not gross, pay.

The size of the efficiency loss is governed by the **elasticity of labor supply** (Lecture 2). Governments can dampen it by using tax revenue to subsidize goods complementary to working, such as child care.

⇒ “How Can Scandinavians Tax So Much?” (Kleven, 2014) — high-tax Nordic countries also invest heavily in making work easy (child care, low-distortion consumption taxes), which is part of how they sustain such high tax revenue with limited efficiency loss.

Source: Kleven, H., *Journal of Economic Perspectives* 28(4), 2014, cited in Leenders, Part IV.

4.3 Measuring the Fairness of Tax Systems

Average vs. Marginal Tax Rates

Marginal tax rate

The share of the *next euro* of taxable income paid in tax.

Average tax rate

Total tax paid, divided by total (gross) income — a weighted average of all the marginal rates paid along the way.

Austria's 2026 Income Tax Brackets

	Tax rate
13.539 und darunter	0 %
über 13.539 bis 21.992	20 %
über 21.992 bis 36.458	30 %
über 36.458 bis 70.365	40 %
über 70.365 bis 104.859	48 %
über 104.859 bis 1.000.000	50 %
über 1.000.000 ¹⁾	55 %

The 55% top bracket is a temporary surcharge. Brackets are progressive: each additional euro is taxed at a higher rate than the last.

Worked Example: Lena's Tax Bill

Lena has taxable income of €80,000. Walking up the bracket schedule:

$$(\text{€}8,453 \times 0.20) + (\text{€}14,466 \times 0.30) + (\text{€}33,907 \times 0.40) + (\text{€}9,635 \times 0.48) = \text{€}24,218.00$$

Her disposable income: $80,000 - 24,218 = 55,782$

- Lena's **marginal** tax rate is **48%** — the rate on her next euro earned.
- Lena's **average** tax rate is $\text{€}24,218.00 / \text{€}80,000 = \mathbf{30.3\%}$.

The gap between the two — 48% vs. 30.3% — is entirely due to progressivity: the first €13,539 was tax-free, and every bracket below her marginal one was taxed at a lower rate.

Vertical and Horizontal Equity

Vertical equity

Groups with more resources should pay higher taxes than groups with fewer resources.

Horizontal equity

Individuals who are similar, but make different choices, should be treated the same by the tax system.

- A **progressive** system: average tax rates *rise* with income.
- A **proportional** (flat) system: average tax rates are *constant*.
- A **regressive** system: average tax rates *fall* with income.

When “Fair” Causes a Riot

In March 1990, rioters in London set fire to parked cars and smashed storefronts, injuring more than 400 people. The cause: Margaret Thatcher’s proposal to replace property taxes with a **poll tax** — a flat charge levied *equally* on every adult, rich or poor.

Why it provoked outrage

The tax burden shifted away from wealthy property owners and onto poorer citizens who hadn’t previously paid property tax at all — a textbook violation of **vertical equity**. Thatcher was ousted as Conservative leader soon after; the tax was abandoned.

(England’s last attempt at such a tax, in 1381, ended with the beheading of several prominent officials.)

The problem

With fixed, progressive brackets, **inflation** pushes people into higher brackets even when their *real* income hasn't grown — a stealth tax increase known as *kalte Progression* (bracket creep).

- Effective 1 January 2023, Austria began **automatically indexing** tax brackets to inflation: two-thirds of the inflation adjustment happens automatically by law each year; the remaining third is allocated at the government's discretion via a separate "*Progressionsabgeltungsgesetz*."
- The top €1,000,000 / 55% bracket is *excluded* from indexation.

4.4 Defining the Income Tax Base

The Haig-Simons Comprehensive Income Definition

Haig-Simons income

Taxable resources = an individual's *ability to pay* = total consumption during the year
+ any increase in wealth.

- In principle, comprehensive income improves *both* vertical equity (all resources are taxed, however received) *and* horizontal equity (the form compensation takes shouldn't matter).
- In practice, every real tax system deviates from this ideal — the question is whether each deviation is justified.

Source: Gruber, *Public Finance and Public Policy*, §18.4.

Austria vs. the US: In-Kind Compensation

- The classic Haig-Simons violation in the US: employer-provided **health insurance** is not taxed as income, even though it clearly raises a worker's ability to consume.
- Austria takes a stricter approach for many in-kind benefits: the *Sachbezugswerteverordnung* taxes the value of a company car, subsidized housing, and other in-kind perks as part of wage income.

Deviations for Ability-to-Pay and Earning Costs

- **Ability-to-pay deviations:** Austria's *außergewöhnliche Belastungen* (extraordinary burden deductions) let taxpayers deduct large, involuntary costs — medical expenses, disaster losses — on the same logic as Gruber's “fire damage” example: these expenditures aren't really *consumption*, so they shouldn't count as ability to pay.
- **Cost-of-earning-income deviations:** Austria's *Werbungskosten* let workers deduct job-related expenses, including the *Pendlerpauschale* (commuter allowance, Lecture 1) — the same logic as deducting business costs from Haig-Simons income.

Both are easy to justify in principle — and hard to draw a clean line around in practice (was that lunch a business cost, or consumption?).

- Whether you rent or own, you consume €X/month of housing services — your *imputed rental value*. A true Haig-Simons tax would treat renters and owners identically.
- The **US** does not tax imputed rental income, *but* lets owners deduct mortgage interest — a large, one-sided subsidy to owning ($\approx$ €80bn/year in forgone US federal revenue)
- Empirical research finds little evidence that home ownership has positive social benefits
Austria taxes owners and renters rather symmetrically

Deductions vs. Credits

Tax deduction

Reduces *taxable income* — its value to the taxpayer is $\$(1 - t)$ per euro, so it's worth *more* to people in higher brackets.

Tax credit

Reduces *tax owed* directly — worth the same to everyone, regardless of bracket, and thus more progressive.

A **refundable** credit goes further still: it pays out even to people who owe little or no tax — turning the tax system into a delivery mechanism for cash transfers to low earners.

Austria's Own Refundability Debate: *Familienbonus Plus*

Introduced in 2019 as a tax **credit** (not a deduction) for families with children.

- 2024 value: €2,000.16/year per child under 18; €700.08/year per child 18 or older.
- Because it's a credit against tax owed, it delivers *nothing* to a parent who owes little or no income tax — exactly the refundability problem the US Child Tax Credit debate raises.
- Austria's partial fix: the *Kindermehrbetrag* (2024: €700/child) tops up low earners who claim the sole-earner/single-parent credit, closing the gap between their (near-zero) tax liability and €700.

Discussion: does the *Kindermehrbetrag* fully solve the vertical-equity problem of a non-refundable credit, or only partly?

Source: Bundeskanzleramt; BMF, Familienbonus Plus & Kindermehrbetrag (2024).

Tax expenditures

Revenue the government forgoes through special exclusions, deductions, credits, or preferential rates — functionally equivalent to direct spending, just routed through the tax code instead of the budget.

In the US, tax expenditures total over €1 trillion/year — more than 40% of total tax revenue. Every deduction and credit on the last several slides (Familienbonus Plus, Werbungskosten, außergewöhnliche Belastungen) is, by this definition, an Austrian tax expenditure too.

Source: Gruber, *Public Finance and Public Policy*, §18.5, citing U.S. Department of the Treasury (2021a).

The Rise of the Fiscal State

How Did We Get Progressive Taxation?

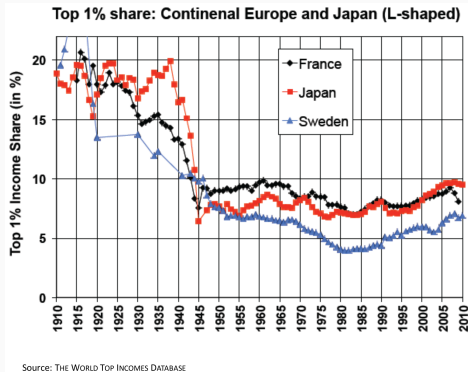
- Top marginal income-tax rates were in the single digits almost everywhere before 1900 — then rose sharply, above all in the US, UK, Sweden, Germany, and France, from the 1910s onward.
- **World War I** acted as the catalyst: many countries raised top rates on income and inheritances explicitly as a “conscription of capital” to match the conscription of soldiers.
- The wartime rate hikes only *stuck* because of lasting technological and institutional changes: employer-based withholding, and — eventually — the end of bank secrecy.

Austria's own version of that last point

Austria was long known for strict *Bankgeheimnis* (bank secrecy). That changed with the OECD's Common Reporting Standard: Austria began automatically exchanging account data with foreign tax authorities starting September 2017.

Source: Leenders, Part IV, citing Piketty, *Capital and Ideology* (2020); Gemeinsamer Meldestandard-Gesetz (GMSG), effective 2016–2017.

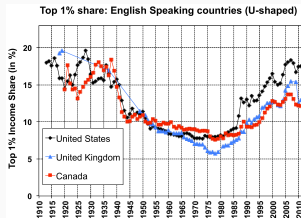
The Long-Run Fall in Top Income Shares: L-Shaped



The rate hikes on the previous slide show up directly in the data: top 1% income shares in France, Japan, and Sweden collapsed from $\approx 18\text{--}20\%$ before WWII to $\approx 4\text{--}8\%$ by the 1970s–80s — and **stayed low** since.

Source: World Top Incomes Database / WID.world.

...vs. U-Shaped: English-Speaking Countries



The US, UK, and Canada saw an equally sharp WWII-era collapse — but then **rebounded** close to pre-war levels by the 2010s.

Why the difference?

Piketty, Saez & Stantcheva (2014) link this directly to *tax policy*: countries that cut top marginal rates the most after 1980 (the US, UK) saw the largest rebound in top income shares; countries that kept rates comparatively high (France, Germany, much of Continental Europe) saw much smaller rebounds.

Source: World Top Incomes Database / WID.world; Piketty, Saez & Stantcheva, “Optimal Taxation of Top Labor Incomes,” *AEJ: Economic Policy* 6(1), 2014.

Why This History Matters

- The rise in top rates on the wealthy over the twentieth century is a major part of the story behind the dramatic *fall* in income and wealth inequality across the same period (Lecture 1).
- Total tax revenue as a share of national income rose alongside it, from roughly 5–10% (1870) to 30–55% (today) across today's rich democracies — with the US consistently at the *low* end of that range and the Nordic countries at the *high* end.

This isn't ancient history — it's the direct backdrop to the debates in this lecture: what should be taxed, how progressively, and how the tax base should be defined.

Source: Leenders, Part IV, citing Piketty (2020) and long-run OECD/WID revenue series.

4.5 The Appropriate Unit of Taxation

Whose Income Do We Tax? Pick Two of Three

Suppose a tax system had to satisfy three goals at once:

- **Progressivity:** marginal tax rates rise as income rises.
- **Across-family horizontal equity:** families with equal total income pay equal tax.
- **Across-marriage horizontal equity:** tax burdens are *marriage neutral* — unaffected by whether two people wed.

All three seem reasonable. It is **impossible** to satisfy all three simultaneously — any progressive tax system taxing couples on some notion of joint income must give up at least one.

Worked Example: Two Couples, Same Income

A stylized progressive schedule: 10% up to €20,000; 20% up to €80,000; 30% above €80,000 (illustrative only — not Austria's actual schedule).

	Individual incomes	Tax, filed individually	Tax, filed as a family
Yasmin & Doug	€140,000 + €10,000	€33,000	€35,000
Jan & Elena	€75,000 + €75,000	€26,000	€35,000

Both couples earn €150,000 combined.

- **Individual filing:** Yasmin & Doug pay €7,000 *more* than Jan & Elena, despite equal family income — violates across-family equity.
- **Family filing:** both couples pay €35,000 — equal treatment, but *both* now pay more than they did unmarried — a **marriage tax**.

Why the Trade-off Is Unavoidable

- Under **individual** filing, rising marginal rates mean two-earner couples with similarly-sized incomes (Jan & Elena) pay less than one-earner-dominant couples (Yasmin & Doug) with the same total — that's the equity violation.
- Under **family** filing, combining incomes pushes the secondary earner's earnings on top of the primary earner's — Doug's first €10,000, taxed at 10% individually, gets taxed at 30% once merged with Yasmin's income. That's the marriage tax.
- The *only* escape from both problems at once is a flat (proportional) tax — which gives up progressivity, the first goal.

Marriage Tax in Austria

- Austria taxes spouses **individually**, not as a family unit — and has done so since 1973, when a family-law reform replaced the previous household-taxation system.
- Because each partner's tax bracket depends only on *their own* income, getting married changes **nobody's** tax bill by itself — Austria's system is marriage-neutral
- The *Alleinverdienerabsetzbetrag* (sole-earner credit) gives a small, means-tested credit to one-earner couples/families with children when the lower-earning partner's income stays below a threshold

The Unit of Taxation Around the World

System	OECD countries
Individual taxation	19 countries, including Austria
Family taxation with income splitting	France, Germany, Luxembourg, Portugal, Switzerland (mostly generates marriage <i>subsidies</i>)
“Pure” family taxation (no splitting)	United States, Ireland, Norway

The Distribution of the Tax Burden

Who Actually Pays the Taxes?

- Rather than asking how a *hypothetical* tax change would affect prices (incidence theory), a newer research tradition (Saez & Zucman, 2019, and successors) asks: given the tax system *as it exists*, what share of everyone's income actually goes to tax, by income rank?
- Applied to the Netherlands, Italy, France, and the US, this research finds effective tax rates that **rise** through the middle of the distribution — then **flatten or fall** at the very top.

The progressivity we build into income-tax *brackets* doesn't automatically show up in the effective rate actually paid, once every tax is counted.

Four Insights on Who Pays

1. **Income taxes** are progressive for the bottom 90% of earners, but play only a minor role at the very top — where income increasingly comes from capital, not wages.
2. **Corporate taxes** act as a kind of minimum tax on the rich, since a large share of top-end income flows through corporate ownership.
3. **Consumption taxes** hit lower-income households hardest, because they consume a larger *share* of their income than the wealthy do.
4. **Payroll taxes** are levied only on labor income — and labor income's share of total income *falls* as you move up the distribution, which is regressive.

Try It Yourself: Design Your Own Tax System

Saez & Zucman built an interactive tool around the same distributional data from the last two slides — it lets you set tax rates by income group and see the revenue and distributional consequences in real time.

<https://taxjusticenow.org/#makeYourOwnTaxPlan>

In-class exercise: pick a distributional goal (e.g. “flat effective rate for everyone” or “restore 1950s-style progressivity at the top”) and see what tax rates it actually takes to get there.

Source: Saez & Zucman, *taxjusticenow.org*, companion site to *The Triumph of Injustice* (2019).

Thank you!

Questions and discussion